

MEDASSIST AI

Medical Symptom Analysis & Disease Prediction System

Project Overview :

Healthcare systems generate large amounts of patient information, including symptoms, medical history, health conditions, and lifestyle information. Analyzing this information manually can require considerable time and expertise. Artificial Intelligence and Machine Learning can be used to analyze healthcare-related data and provide preliminary insights.

MedAssist AI is designed as a centralized healthcare intelligence platform that connects symptom analysis, disease prediction, risk assessment, healthcare recommendations, reporting, and analytics.

The system follows a workflow where the user provides symptoms, the AI engine analyzes the information, possible diseases are predicted, health risks are evaluated, and suitable preliminary recommendations are generated.

The platform can support different healthcare stakeholders such as patients, healthcare providers, clinics, hospitals, telemedicine platforms, and healthcare organizations.

Objectives :

The major objectives of MedAssist AI are:

1. To develop an AI-powered medical symptom analysis system.
2. To predict possible diseases based on user symptoms.
3. To calculate and categorize health risks.
4. To provide preliminary healthcare recommendations.
5. To implement secure user authentication.
6. To provide role-based access control.
7. To maintain patient health history.
8. To generate downloadable healthcare reports.
9. To provide analytics and health trend visualization.
10. To support healthcare providers through patient report review.

The intended outcomes include authentication, symptom analysis, disease prediction, risk assessment, treatment recommendations, health reports, patient history, and healthcare analytics.

1.Problem Statement

Patients commonly experience symptoms without knowing whether they indicate a minor condition or a potentially serious health problem. Traditional healthcare processes generally require patients to consult healthcare professionals before receiving an assessment.

Although professional medical consultation is essential, an intelligent preliminary symptom-analysis system can help users organize their symptoms and understand potential health conditions before consultation.

The major problems addressed by this project are:

- Difficulty in understanding multiple symptoms.
- Lack of immediate preliminary health insights.
- Manual analysis of symptom combinations.
- Difficulty in identifying potential health risks.
- Lack of centralized patient health information.
- Limited access to automated healthcare recommendations.
- Difficulty in monitoring health trends.
- Need for structured healthcare reports.

Project Scope

The scope of MedAssist AI includes:

Patient Scope

- User registration and login.
- Patient profile management.
- Symptom submission.
- Disease prediction.
- Risk assessment.
- Healthcare recommendations.
- Health report generation.
- Patient history management.

Healthcare Provider Scope

- Patient information review.
- Prediction report review.

- Clinical notes.
- Report approval or rejection.

Administrative Scope

- User management.
- System monitoring.
- Audit and activity management.

Analytics Scope

- Disease prediction analytics.
- Symptom trend analysis.
- Health insights.
- System performance monitoring.

System Architecture

MedAssist AI follows a layered architecture consisting of frontend, backend, database, machine learning, reporting, and deployment components.

Architecture Layers

Presentation Layer

The frontend provides the user interface through which patients, doctors, clinics, and administrators interact with the platform.

The frontend handles:

- Login
- Registration
- Dashboard
- Symptom selection
- Prediction results
- Risk assessment
- Recommendations
- Reports
- Analytics

2.Application Layer

The FastAPI backend provides REST APIs for communication between the frontend and system services.

It handles:

- Authentication
- User management
- Symptoms
- Predictions
- Risk assessment
- Recommendations
- Reports
- Analytics

AI/ML Layer

The AI engine processes the symptom information and generates disease predictions with probability/confidence information. Database Layer

The application uses databases to store user information, patient information, healthcare records, and system activity.

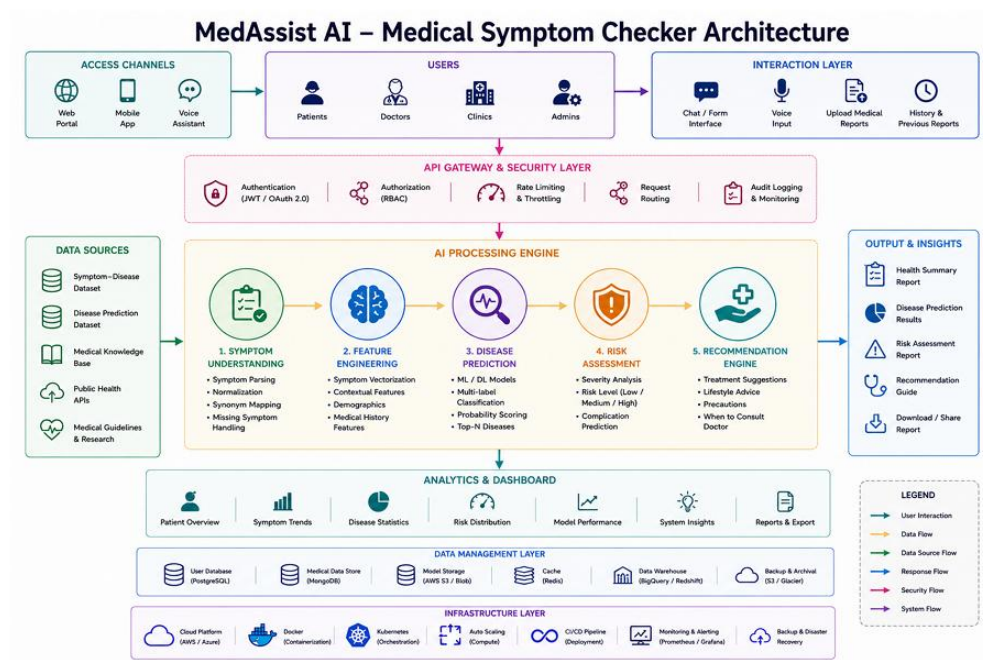
Reporting Layer

The reporting engine generates downloadable health reports containing prediction, risk, and recommendation information.

Deployment Layer

Docker and Docker Compose are used to provide reproducible application deployment.

3. Architecture Diagram



Technology Stack

Frontend Technologies

Next.js:

Next.js is used to build the web application interface and provide structured frontend development.

React :

React is used for creating reusable user interface components.

Tailwind CSS:

Tailwind CSS is used for responsive styling and dashboard design.

Backend Technologies :

Python:

Python is used as the primary programming language for backend and AI-related functionality.

FastAPI:

FastAPI provides REST APIs for communication between frontend and backend services.

3.Database Technologies

MySQL

MySQL is used for structured application data.

MongoDB

MongoDB is used for suitable document-based information and logging-related functionality.

AI/ML Technologies

- Scikit-learn
- Machine Learning classification
- Random Forest
- Pandas
- NumPy

Security Technologies

- JWT authentication
- Password hashing
- Role-Based Access Control

DevOps & Deployment

- Docker
- Docker Compose
- Git
- GitHub
- Postman The project specification also identifies FastAPI, React/Next.js, Tailwind CSS, JWT, Scikit-learn, TensorFlow/XGBoost, Chart.js, Docker, AWS/Azure, GitHub, and Postman among the technology ecosystem.

4. System Modules

User Management Module:

The User Management Module provides registration, login, profile management, role management, and medical history functionality.

Symptom Analysis Module:

This module allows users to select or enter symptoms. The system validates the symptoms and prepares the information for disease prediction.

Disease Prediction Module:

The Disease Prediction Module analyzes the symptom vector and predicts possible diseases. The system can generate multiple possible disease predictions with probability/confidence information.

Risk Assessment Module:

The Risk Assessment Module evaluates the severity of the predicted condition and categorizes health risks.

Recommendation Module:

The recommendation engine provides:

- Healthcare recommendations
- Preventive-care suggestions
- Lifestyle guidance
- Follow-up guidance

Health Reports Module

The report module generates:

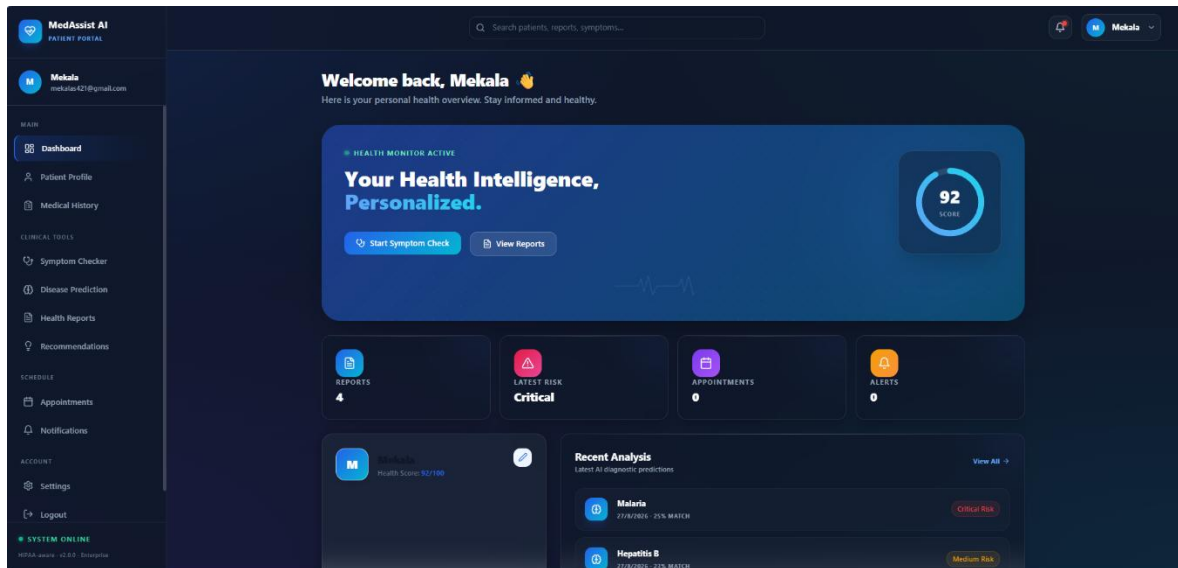
- Disease prediction reports
- Risk assessment reports
- Patient health summaries
- Downloadable reports

Analytics Dashboard

The analytics module provides:

- Disease prediction analytics
- Symptom trends
- Healthcare insights
- System performance monitoring

These core modules are directly aligned with the module structure in the project demo specification.



4.Dataset & Data Preprocessing

Dataset:

Healthcare datasets are required to train and evaluate disease prediction and risk assessment systems.

The project specification identifies the following datasets:

Disease Symptoms and Patient Profile Dataset

Purpose:

- Symptom-to-disease prediction
- Disease classification
- Medical recommendation workflows

Example disease categories include:

- Diabetes
- Malaria
- Dengue
- Hypertension
- Asthma

5.Disease Prediction Using Symptoms Dataset

This dataset supports:

- Multi-class disease prediction
- Symptom correlation analysis
- Symptom combination analysis

CDC BRFSS Dataset

The CDC Behavioral Risk Factor Surveillance System dataset can support:

- Health risk assessment
- Population health analytics
- Lifestyle-based health analysis

MIMIC-IV

MIMIC-IV is identified as an optional advanced dataset for:

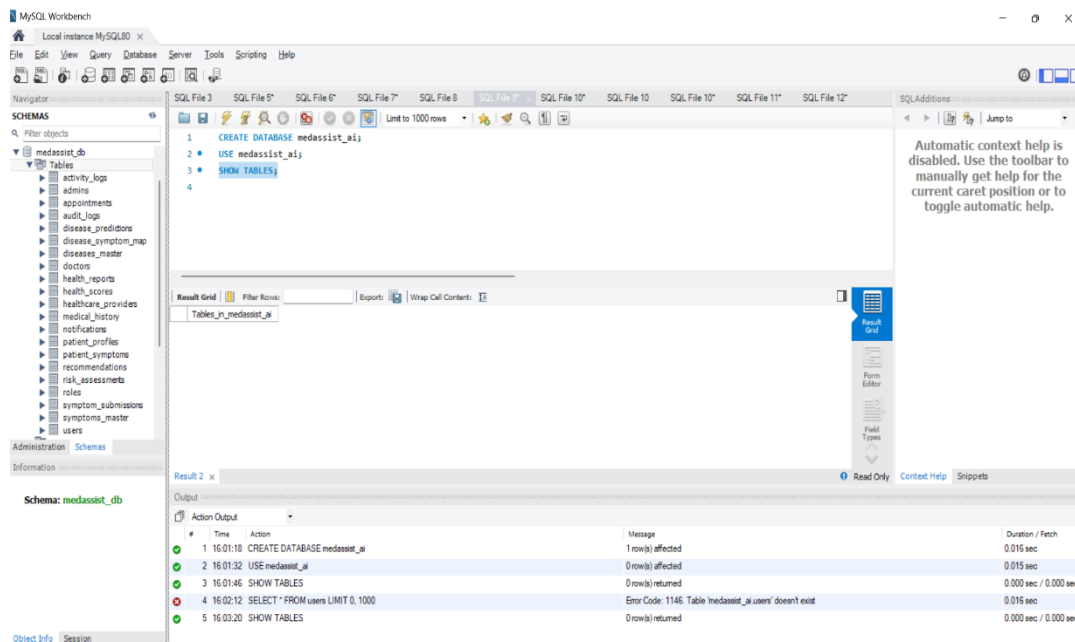
- Clinical decision support
- Healthcare analytics
- Patient risk prediction

The demo specification recommends the Disease Symptoms and Patient Profile Dataset as the primary student-level dataset because of its disease-symptom mapping and suitability for classification.

6.Data Preprocessing

Typical preprocessing workflow includes:

1. Loading the dataset.
2. Removing invalid records.
3. Handling missing values.
4. Standardizing symptom names.
5. Encoding disease labels.
6. Creating symptom vectors.
7. Splitting data for model development.
8. Validating the processed dataset.



7.AI/ML Model

Disease Prediction

The AI component is responsible for identifying possible diseases based on patient symptoms.

The prediction process can be represented as:

Patient Symptoms → Symptom Vector → ML Classifier → Disease Prediction → Probability/Confidence → Risk Assessment

Random Forest Classifier

Random Forest is used as the primary classification approach in the implemented prediction pipeline.

A Random Forest model consists of multiple decision trees. Each tree independently produces a prediction, and the combined results are used to generate the final classification.

The model is suitable for symptom-based classification because multiple symptoms can contribute to the prediction of different disease classes.

Prediction Output

The system generates:

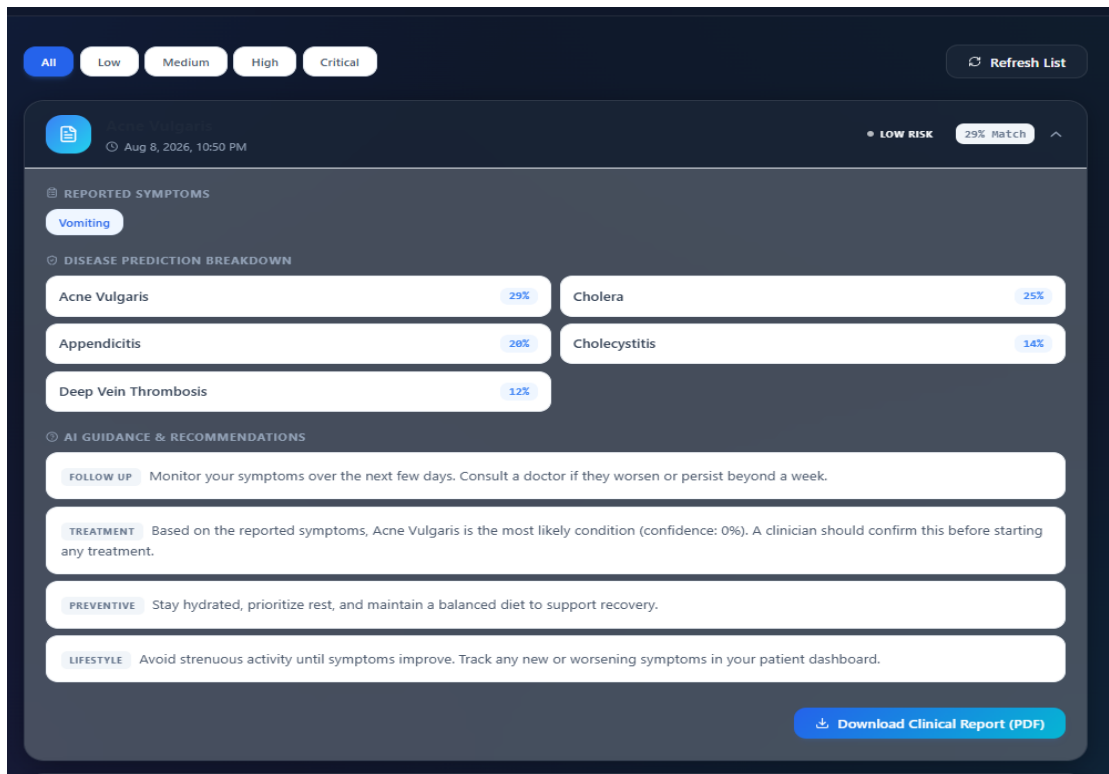
- Predicted disease
- Multiple possible conditions
- Probability information
- Confidence information

Fallback Prediction

For unsupported or unavailable model situations, the system can use a symptom-overlap approach to compare submitted symptoms with disease-symptom relationships.

Risk Assessment

The prediction output is passed to the risk assessment component, which evaluates severity and categorizes the patient's health risk.



8. Implementation

The implementation of MedAssist AI follows an end-to-end healthcare workflow.

User Registration

Users create an account by providing required information.

Authentication

After registration, users authenticate using their credentials. JWT-based authentication is used to maintain secure sessions.

Symptom Collection

The patient selects or enters symptoms through the symptom analysis interface.

AI Prediction

The backend processes the symptoms and sends them to the AI prediction engine.

Risk Assessment

The predicted condition and symptom information are evaluated to determine the health risk level.

Recommendation

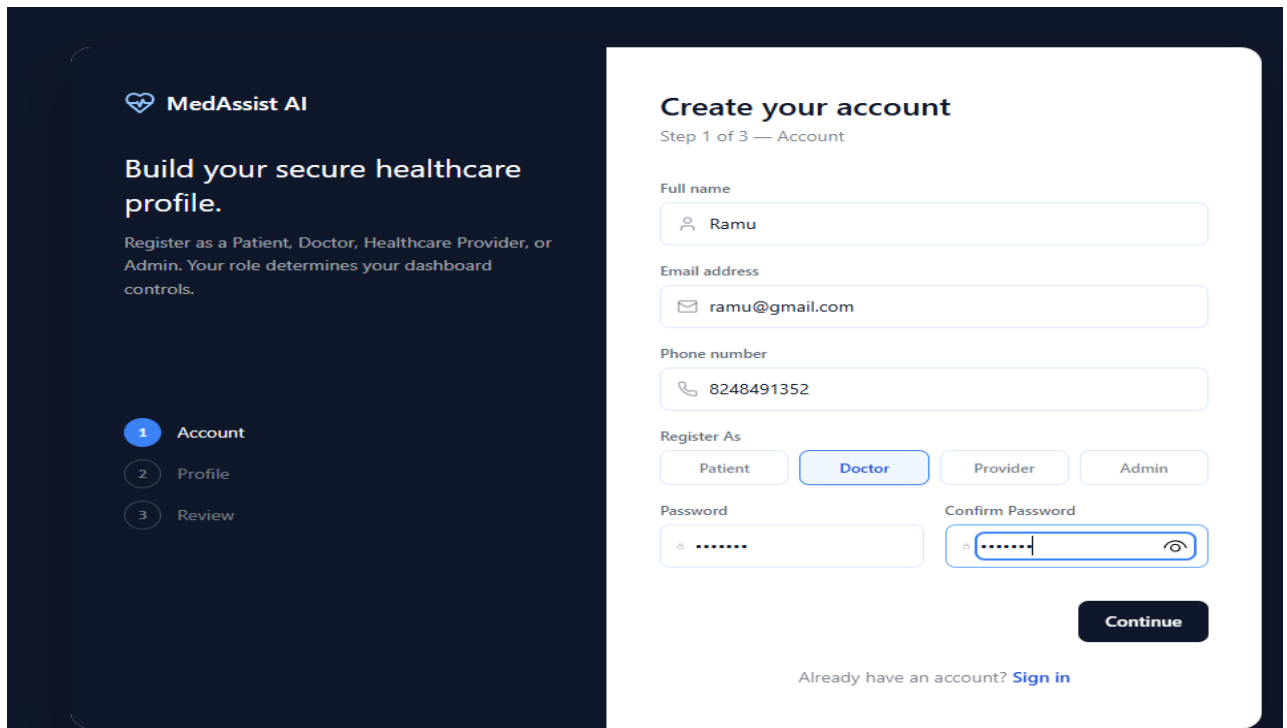
The system generates preliminary healthcare and preventive-care recommendations.

Doctor Review

Healthcare providers can review the prediction and patient information and provide clinical notes where applicable.

Report Generation

The system generates a structured health report containing prediction, risk, and recommendation information.



The image shows a user interface for creating an account with MedAssist AI. On the left, a dark sidebar contains the MedAssist AI logo and the heading "Build your secure healthcare profile." Below this, it says "Register as a Patient, Doctor, Healthcare Provider, or Admin. Your role determines your dashboard controls." A vertical list of steps is shown: 1 Account (highlighted), 2 Profile, and 3 Review. The main area is titled "Create your account" and "Step 1 of 3 — Account". It contains several input fields: "Full name" with the value "Ramu", "Email address" with "ramu@gmail.com", and "Phone number" with "8248491352". There is a "Register As" section with four buttons: "Patient", "Doctor" (which is highlighted in blue), "Provider", and "Admin". Below this are "Password" and "Confirm Password" fields, both containing masked characters. The "Confirm Password" field has an eye icon to toggle visibility. A dark blue "Continue" button is at the bottom right. At the very bottom, there is a link: "Already have an account? Sign in".

MedAssist AI

Build your secure healthcare profile.

Register as a Patient, Doctor, Healthcare Provider, or Admin. Your role determines your dashboard controls.

1 Account
2 Profile
3 Review

Create your account
Step 1 of 3 — Account

Full name
Ramu

Email address
ramu@gmail.com

Phone number
8248491352

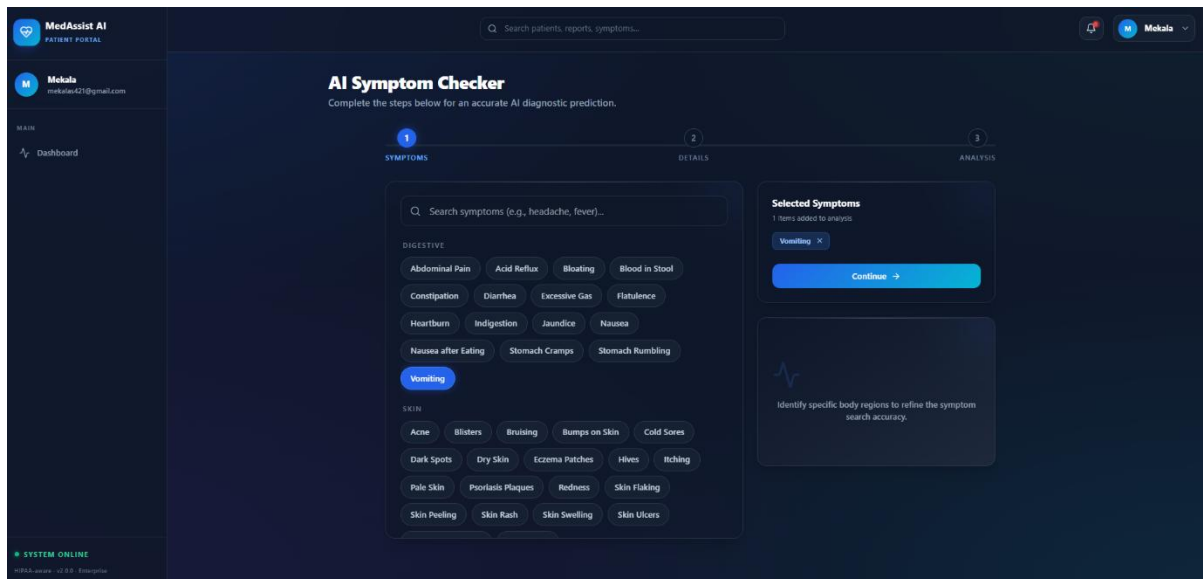
Register As
Patient Doctor Provider Admin

Password
.....

Confirm Password
.....

Continue

Already have an account? [Sign in](#)



9. Implementation & Milestones

Milestone 1 – Project Initialization

During Weeks 1 and 2:

- Project objectives were defined.
- Healthcare workflows were planned.
- System architecture was designed.
- Database structure was planned.
- Authentication was implemented.
- Patient management was developed.
- Symptom collection workflow was established.

Milestone 2 – Disease Prediction & Risk Assessment

During Weeks 3 and 4:

- Disease prediction model was developed.
- Model performance was evaluated.
- Probability scoring was implemented.
- Risk scoring was developed.
- Severity analysis was implemented.
- Health risk reports were generated.

Milestone 3 – Recommendation & Analytics

During Weeks 5 and 6:

- Recommendation workflows were developed.
- Treatment and preventive suggestions were implemented.
- Analytics dashboard was developed.
- Disease prediction reports were generated.
- Health trend visualization was implemented.

Milestone 4 – Testing & Deployment

During Weeks 7 and 8:

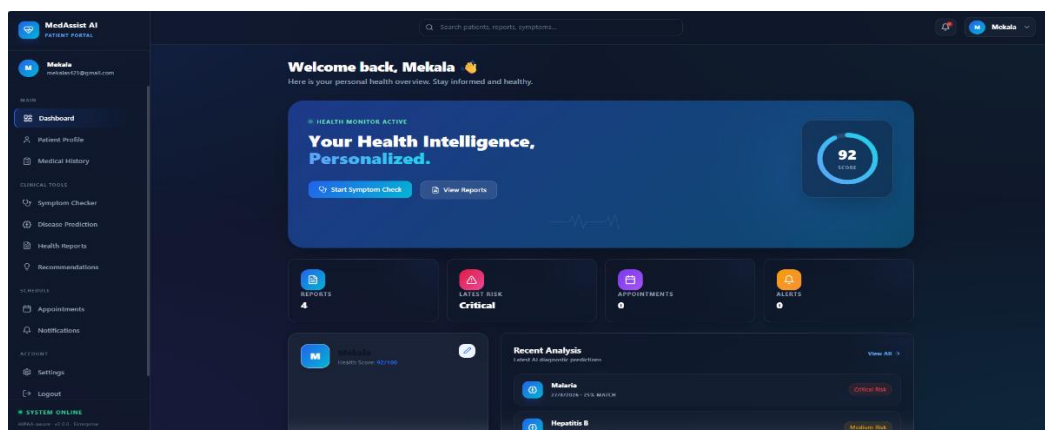
- Model validation was performed.
- System workflows were tested.
- Dashboard responsiveness was evaluated.
- Docker deployment was prepared.
- Documentation and final demonstration were completed.

This milestone progression follows the implementation schedule specified in the demo document.

10. User Roles & Dashboard

MedAssist AI supports different user roles with role-based access.

- **Patient:** Manage profile, enter symptoms, view predictions, risk, recommendations and reports.
- **Doctor:** Review patient information, prediction reports and add clinical notes.
- **Clinic:** Manage patients and healthcare workflows.
- **Admin:** Manage users and monitor system activities.

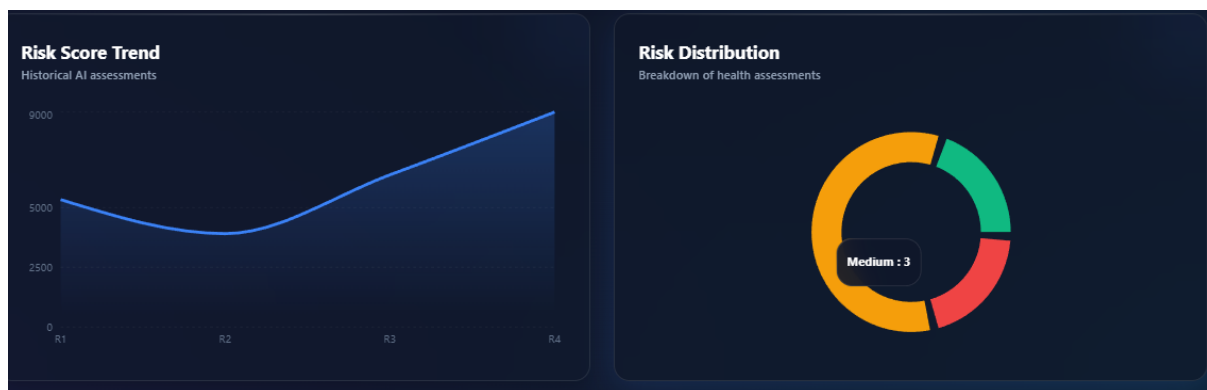


11. Dashboard & Analytics

The dashboard provides a centralized view of healthcare information.

- Patient health summary
- Recent symptoms and predictions
- Risk information
- Healthcare recommendations
- Reports and history
- Disease and symptom analytics
- Health trend visualization

The analytics dashboard helps users understand disease patterns and healthcare insights.



12. Testing & Performance Evaluation

The system is tested to verify functionality, reliability and security.

Major testing areas include:

- User authentication testing
- Role-based access testing
- Symptom submission testing
- AI prediction testing
- Risk assessment testing
- Recommendation testing
- PDF report testing
- Analytics testing
- API performance testing

13. Performance Evaluation

The system performance was evaluated using response-time measurements.

Operation	Response Time
Login	219.06 ms
Symptoms List	13.91 ms
AI Prediction	127.31 ms
PDF Generation	367.70 ms
Analytics	20.85 ms

Important evaluation metrics include accuracy, precision, recall, F1-score, prediction confidence and API response time.

14. Deployment

MedAssist AI uses containerized deployment with Docker and Docker Compose.

The application contains:

- Frontend
- Backend
- MySQL
- MongoDB

Docker provides a consistent and reproducible environment for development and deployment. The project specification also identifies AWS/Azure as possible cloud deployment platforms.

15. Results

The implemented system successfully provides an end-to-end healthcare workflow.

Major Results

- User authentication
- Symptom analysis
- Disease prediction
- Risk assessment
- Healthcare recommendations
- Doctor review

- Health report generation
- Analytics dashboard
- PDF report generation

The system provides preliminary healthcare insights based on user-provided symptoms and supports informed healthcare decisions.

MedAssist AI

Clinical Decision Support Report

Patient Name: Mekala
Email: mekalas421@gmail.com
Patient ID: ab30fb81-34bc-48c7-9671-0eb9375696a2
Report Date: 08/08/2026, 22:02:43
Report ID: 9be98f0d-ccdf-4246-8b5d-5ed982cdf3f8

Diagnostic Overview

Primary AI Diagnosis: Acne Vulgaris
Confidence Score: 36%
Risk Level: LOW

Reported Symptoms

Reported Symptom	Severity
Acid Reflux	Unknown

Differential Diagnosis

Disease Name	Probability
Acne Vulgaris	36%
GERD	27%
Fibroids	15%
Deep Vein Thrombosis	12%
ADHD	8%

AI Guidance & Recommendations

Recommendation Type	Guidance
FOLLOW UP	Monitor your symptoms over the next few days. Consult a doctor if they worsen or persist beyond a week.
TREATMENT	Based on the reported symptoms, Acne Vulgaris is the most likely condition (confidence: 0%). A clinician should confirm this before starting any treatment.
PREVENTIVE	Stay hydrated, prioritize rest, and maintain a balanced diet to support recovery.
LIFESTYLE	Avoid strenuous activity until symptoms improve. Track any new or worsening symptoms in your patient dashboard.

16. Conclusion

MedAssist AI integrates Artificial Intelligence, Machine Learning and modern web technologies to provide healthcare assistance.

The system combines:

- Symptom analysis
- Disease prediction
- Risk assessment
- Recommendations

- Doctor review
- Health reports
- Analytics

The project demonstrates how AI can support preliminary healthcare assessment and help users understand potential health conditions.

17. Future Enhancements

Future versions of MedAssist AI can include:

- **EHR Integration** using FHIR and HL7
- **Multi-language Support**
- **Telemedicine Integration**
- Advanced AI/ML models
- Explainable AI
- Mobile application
- Wearable-device health monitoring

EHR, multilingual support and telemedicine are specifically identified as future directions in the project specification.

18. References

- Disease Symptoms and Patient Profile Dataset
- Disease Prediction Using Symptoms Dataset
- CDC BRFSS Dataset
- MIMIC-IV Dataset
- Python
- FastAPI
- Next.js
- React
- Scikit-learn
- MySQL
- MongoDB
- Docker
- Git & GitHub